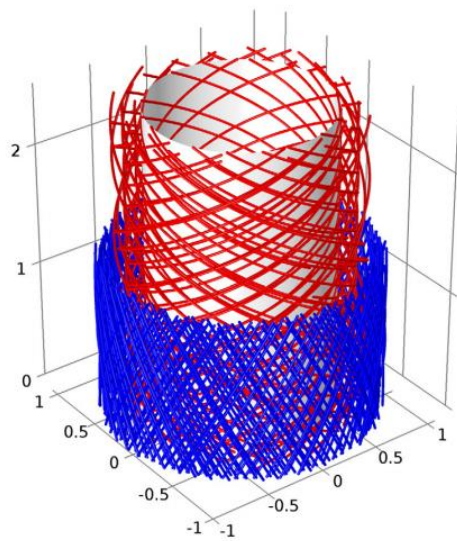


Arterial Wall Mechanics

A review of Constitutive Models and Implementation in COMSOL Multiphysics



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1.0 Introduction

The increasing interest in modeling arterial wall mechanics stemmed from Humphrey's comparative study on constitutive models to find the mechanical response of arteries. An explanation for this increase is the realization of mechanical forces influencing diseases such as atherosclerosis which has led to high mortality rates in the western world. To understand these mechanical forces, we require accurate constitutive models which describe how arterial tissue deforms under different loads.

Arteries have very complex structures and experience internal blood pressure and stresses that come from natural growth and remodeling of themselves. The authors show that normal models don't consider these features which affect their predictive abilities and present their new model which introduces collagen fibers to the structure of arteries.

The motivation behind a new and reliable model is to understand diseases such as atherosclerosis and capture the complex behavior of arteries under deformation and further improve simulations for clinical procedures.

2.0 Structural Properties of Arterial Walls

Understanding these challenging models requires an understanding of the background and structure of arterial walls. Arteries are composed of 3 layers: Intima, Media, and Adventitia. As shown in Figure 1, these 3 layers have their own composition and play their own mechanical functions in the models.

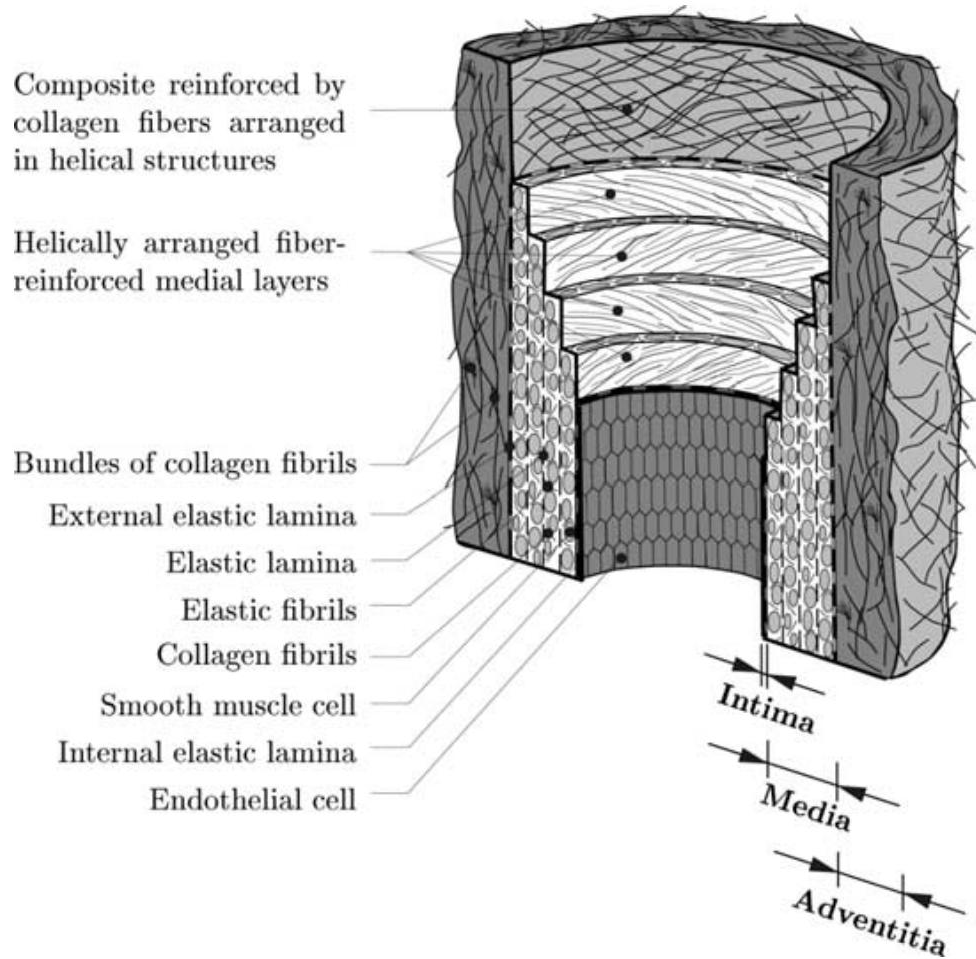


Figure 1: Model of the major components of a healthy elastic artery composed of 3 layers

2.1 Intima

Intima is the thin inner layer of the artery. Resting on a thin basal membrane, the intima is mostly made up of endothelial cells. Its role becomes important in conditions such as atherosclerosis for older, less healthy individuals where plaques accumulate in their arteries leading to a buildup and creating a stiff layer. For healthy and young individuals, it has minimal mechanical contributions.

2.2 Media

The media is the middle layer of the artery, and it plays an important role in the mechanical function. It is made up of collagen, elastin, and smooth muscle cells which are formed into a structurally complex layer. The elastin fibers help protect and restore the artery to its original shape after experiencing deformations. The collagen fibers on the other hand are what further stiffen the arteries after large strains. The media has this helical shaped fiber structure composed of reinforced fiber layers, which gives the artery its strength to handle different kinds of stresses.

2.3 Adventitia

Adventitia is the last outer layer of the artery, made up of primarily collagen fibers in a helical structure and loose connective tissue. It plays an important role in high pressure because it provides stiffness to the arteries, even though it is less stiff than the media at low pressures. The collagen fibers are influenced by the pressure on the arteries. For example, at high pressures the fibers straighten out and make the artery stiffer which helps stop it from tearing. Therefore, the mechanical aspect of the adventitia becomes important when the artery is under deformation.

Together these 3 layers form a nonlinear and anisotropic composite structure that can experience different kinds of forces.

3.0 Mechanical Behavior of Arterial Tissue

3.1 Components of Mechanical Behavior

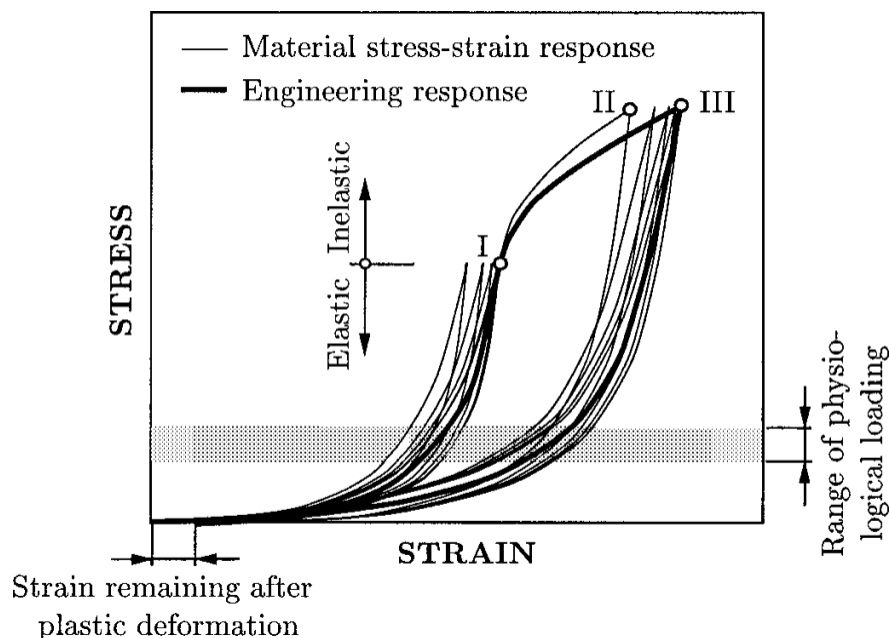
Elastin: dominates at low strains; provides elasticity.

Collagen: recruited at higher strains; provides stiffness (especially in the adventitia).

Ground substance and smooth muscle cells: contribute to viscoelastic and active behavior.

3.2 Typical Mechanical Behavior

Arteries show nonlinear stress-strain curves with exponential stiffening under high pressures. Figure 2 shows this stress-strain response with a circumferential arterial strip, located in the media, under uniaxial tension.



Initially we notice a low stiffness region, during the cyclical loading and unloading where we see less resistance. Then as more collagen fibers are introduced, we see an elastic and exponential

increase in the curve as the tissue reaches point 1. When the tissue gets stretched to point 2 it experiences this inelastic deformation and won't return to its original state. It then stabilizes at point 3 which behaves like point 1, and the black line shows the general mechanical response of the tissue.

4.0 Finite Hyperelasticity

This part of the paper shows the mathematical principles used to describe the behavior of the artery subject to different kinds of loadings. Arteries are expected to behave like hyperelastic materials after large deformations and a way to show this can be derived from a strain energy function. The determinant $J = \det(\mathbf{F})$ is important for measuring the volume change during the large deformations. Then to modify the tensor to isolate the volume change from the shape change, the deformation gradient $\mathbf{F}$, is decomposed. This helps us use the modified Cauchy Green tensor $\mathbf{C}$ in eq (2), making it independent to volume changes.

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = J^{2/3} \bar{\mathbf{C}}, \quad \bar{\mathbf{C}} = \bar{\mathbf{F}}^T \bar{\mathbf{F}}, \quad (2)$$

4.1 Hyperelasticity Stress Response

To describe stress in hyperelastic materials, the Helmholtz free energy function introduces important elements such as strain and structural tensors.

$$\Psi(\mathbf{E}, \mathbf{A}_1, \dots, \mathbf{A}_n) = U(J) + \bar{\Psi}(\bar{\mathbf{E}}, \mathbf{A}_1, \dots, \mathbf{A}_n), \quad (5)$$

In equation 5 we see U which represents volumetric response and the function Ψ represents the isochoric part.

The 2nd Piola- Kirchhoff stress tensor is introduced in eq (6):

$$\mathbf{S} = \mathbf{S}_{\text{vol}} + \bar{\mathbf{S}} \quad \text{with } \mathbf{S}_{\text{vol}} = \frac{\partial U(J)}{\partial \mathbf{E}}, \quad \bar{\mathbf{S}} = \frac{\partial \bar{\Psi}(\bar{\mathbf{E}}, \mathbf{A}_1, \dots, \mathbf{A}_n)}{\partial \mathbf{E}}. \quad (6)$$

The equation shows us how internal stress connects to the energy that is stored during deformation. We see repeat variables from previous equations as we are introduced to new variables such as $\mathbf{S}$ and $\mathbf{E}$. $\mathbf{S}$ represents the Second Piola–Kirchhoff stress tensor while the $\mathbf{E}$ is the Green–Lagrange strain tensor. The tensor becomes defined with respect to the reference configuration.

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{\text{vol}} + \bar{\boldsymbol{\sigma}} \quad \text{with } \boldsymbol{\sigma}_{\text{vol}} = p\mathbf{I}, \quad \bar{\boldsymbol{\sigma}} = J^{-1} \text{dev} \left(\bar{\mathbf{F}} \frac{\partial \Psi}{\partial \bar{\mathbf{E}}} \bar{\mathbf{F}}^T \right), \quad (10)$$

Then we are introduced to the alternative form of the equation for the Cauchy stress in the deformed configuration. Eq 10 is the result from the deformation gradient being used while transforming the Piola-Kirchhoff stress (restated from eq, 8). The two parts of the equation involve hydrostatic pressure, a Lagrange multiplier when introduced as variable p , and then the fiber structure as it accounts for stretching and torsion. The operator $\text{dev}(\cdot)$ is the deviatoric operator and it removes the isotropic part of a tensor to focus on distortional stress, which is useful for shape changing tissue.

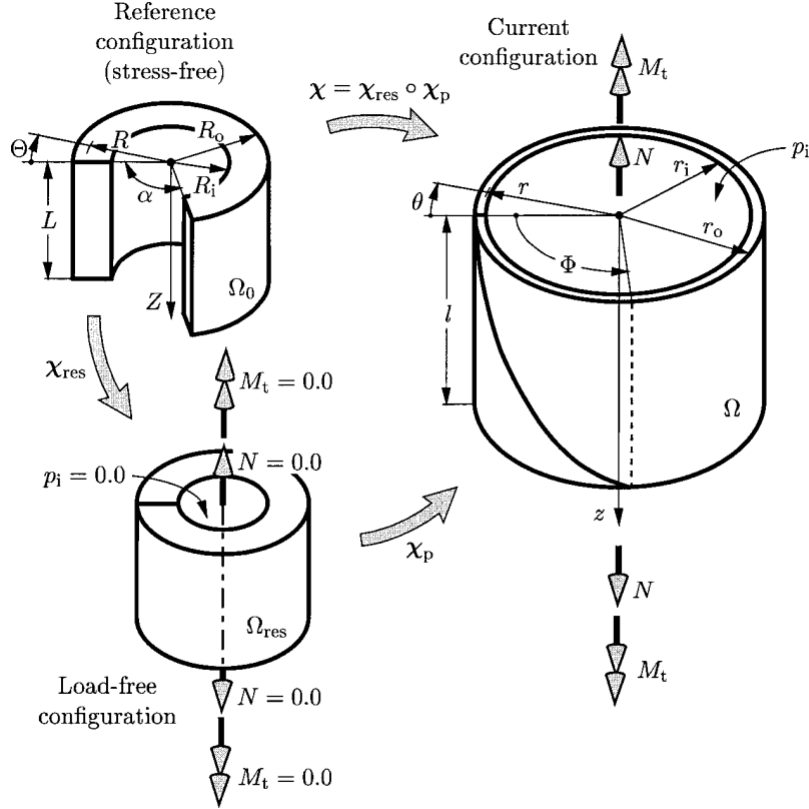


Figure 3. Arterial ring in the the (stress-free) reference configuration Ω_0 , the load-free configuration Ω_{res} and the current configuration Ω .

This figure shows how the arterial model starts in the stress-free and opened configuration and then when placed under bending it goes into the closed shape in the load free configuration after introducing residual stress. Further deformation such as stretching moves the artery into the final state. From this, we can see how stress changes through each step, from internal structure to residual stress and then to external loads which are shown by the equations above.

4.2 Basic Kinematics

The basic kinematics equations help us understand how arteries are influenced by mechanical loads. To figure this out, we look at how arteries deform in geometry or in motion as modeled in the article as a cylindrical tube. In its stress-free or reference configuration, the

deformations are measured with the help of the following reference geometry shown in equation 12:

$$R_i \leq R \leq R_o, \quad 0 \leq \Theta \leq (2\pi - \alpha), \quad 0 \leq Z \leq L, \quad (12)$$

Here R_i , R_o , α and L represent the inner and outer radius. This configuration helps us understand how the artery would behave in its open state or if it was cut open. To show how the artery would behave under loads in its functional state, the article maps the reference to deformed configuration that accounts for loads such as bending or torsion which is shown in equation 15.

$$r = \sqrt{\frac{R^2 - R_i^2}{k\lambda_z} + r_i^2}, \quad \theta = k\Theta + Z\frac{\Phi}{L}, \quad z = \lambda_z Z, \quad (15)$$

In this equation, r , θ , and z are the deformed coordinates, while k represents the arterial wall going from an open to closed shape. λ is the axial stretch ratio, and Φ represents the angle of twist applied over the length L in equations 15(2,3). Then from the mapping of its functioning state, the arterial stretching in each direction needs to be calculated, shown in the stretch equations below:

$$\begin{aligned} \lambda_r(R) &= \frac{\partial r}{\partial R} = \frac{R}{rk\lambda_z}, & \lambda_\theta(R) &= \frac{r}{R} \frac{\partial \theta}{\partial \Theta} = \frac{kr}{R}, \\ \gamma(R) &= r \frac{\partial \theta}{\partial z} = r \frac{\Phi}{L}. \end{aligned} \quad (16)$$

In equation 16 we see the radial, circumference stretching, and shear strain from torsion. These values are used to define the deformation gradient F , which shows the combination of stretching and shear in the arteries deformation.

$$\mathbf{F} = \bar{\mathbf{F}} = (\lambda_\theta \lambda_z)^{-1} \mathbf{e}_r \otimes \mathbf{E}_R + \lambda_\theta \mathbf{e}_\theta \otimes \mathbf{E}_\Theta + \gamma \lambda_z \mathbf{e}_\theta \otimes \mathbf{E}_Z + \lambda_z \mathbf{e}_z \otimes \mathbf{E}_Z, \quad (19)$$

The tensor shown in equation 19 repeats the previously stated stretches in the various directions through the e components. It is useful for calculating strain and stress tensors. Finally in

equations 20 and 21, we can understand where the right and left Cauchy-green deformation tensors were constructed from.

$$\begin{aligned} \mathbf{C} = \overline{\mathbf{C}} = & (\lambda_\theta \lambda_z)^{-2} \mathbf{E}_R \otimes \mathbf{E}_R + \lambda_\theta^2 \mathbf{E}_\Theta \otimes \mathbf{E}_\Theta + \lambda_z^2 (1 + \gamma^2) \mathbf{E}_Z \otimes \mathbf{E}_Z \\ & + \gamma \lambda_\theta \lambda_z (\mathbf{E}_\Theta \otimes \mathbf{E}_Z + \mathbf{E}_Z \otimes \mathbf{E}_\Theta), \end{aligned} \quad (20)$$

$$\begin{aligned} \mathbf{b} = \overline{\mathbf{b}} = & (\lambda_\theta \lambda_z)^{-2} \mathbf{e}_r \otimes \mathbf{e}_r + (\lambda_\theta^2 + \gamma^2 \lambda_z^2) \mathbf{e}_\theta \otimes \mathbf{e}_\theta + \lambda_z^2 \mathbf{e}_z \otimes \mathbf{e}_z \\ & + \gamma \lambda_z^2 (\mathbf{e}_\theta \otimes \mathbf{e}_z + \mathbf{e}_z \otimes \mathbf{e}_\theta). \end{aligned} \quad (21)$$

These tensors show the geometric deformations that are needed to calculate stress in hyperelastic materials. From the reference configuration we use the strain tensors to understand the strain energy function.

4.3 Equilibrium Equations

After the mathematical equations showing how an artery deforms from its reference configuration, the authors explain how internal stresses balance the external loads such as stretching or pressure. This is shown by equilibrium equations which make sure that the artery stays stable under these conditions. In equation 22 we are introduced to the equilibrium equation for a solid at rest under no forces.

$$\text{div } \boldsymbol{\sigma} = \mathbf{0}, \quad (22)$$

This equation makes sure that there aren't any internal forces in play on the artery. In equation 23 we see how in cylindrical coordinate system, it simplifies to a single equation in the r direction.

$$\frac{d\sigma_{rr}}{dr} + \frac{(\sigma_{rr} - \sigma_{\theta\theta})}{r} = 0 \quad (23)$$

Equation 23 shows the importance of balancing stress, and it sets up for calculating pressure and wall stresses in the system. To solve radial stresses, equation 24 takes the previous equation and fixes it from the outwards surface to the internal.

$$\sigma_{rr}(\xi) = \int_{\xi}^{r_o} (\sigma_{rr} - \sigma_{\theta\theta}) \frac{dr}{r}, \quad r_i \leq \xi \leq r_o. \quad (24)$$

Assuming that the outer surface is stress free, the equation for the internal pressure is simplified using the isochoric components of stress, which gives us the required pressure to have the artery in its deformed shape. The authors also calculate the torsional moment in equation 27(2), and the axial force (28), by integrating stress components over the arterial wall.

$$p_i = \int_{r_i}^{r_o} (\bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) \frac{dr}{r}, \quad M_t = 2\pi \int_{r_i}^{r_o} \bar{\sigma}_{\theta z} r^2 dr, \quad (27)$$

$$N = 2\pi \int_{r_i}^{r_o} \left[\int_{r_o}^{\xi} (\bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) \frac{dr}{r} - \bar{\sigma}_{rr} + \bar{\sigma}_{zz} \right] \xi d\xi. \quad (28)$$

Finally, the axial force which is then adjusted for internal pressure is shown in equation 29. This is useful for their inflation testing as it shows accurately the extra axial force being implemented beyond internal pressure.

$$F = \pi \int_{r_i}^{r_o} (2\bar{\sigma}_{zz} - \bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) r dr \quad (29)$$

